

Vitamin A and Carotenoids

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Introduction

Vitamin A is the name of a group of fat-soluble retinoids, primarily retinol and retinyl esters [1,2]. Vitamin A is involved in immune function, cellular communication, growth and development, and male and female reproduction [1-3]. Vitamin A supports cell growth and differentiation, playing a critical role in the normal formation and maintenance of the heart, lungs, eyes, and other organs [1,2]. Vitamin A is also critical for vision as an essential component of rhodopsin, the light-sensitive protein in the retina that responds to light entering the eye, and because it supports the normal differentiation and functioning of the conjunctival membranes and cornea [2,4].

The human diet contains two sources for vitamin A: preformed vitamin A (retinol and retinyl esters) and provitamin A carotenoids [1,5]. Preformed vitamin A is found in foods from animal sources, including dairy products, eggs, fish, and organ meats [1,2]. Provitamin A carotenoids are plant pigments that include beta-carotene, alpha-carotene, and beta-cryptoxanthin [1]. The body converts provitamin A carotenoids into vitamin A in the intestine via the beta-carotene monooxygenase type 1 BCMO1 enzyme [1,3,6], although conversion rates may have genetic variability [7,8,9]. Other carotenoids in food, such as lycopene, lutein, and zeaxanthin, are not converted into vitamin A and are referred to as non-provitamin

A carotenoids; they might have other important activities not involving vitamin A formation [1].

The various forms of vitamin A are solubilized into micelles in the intestinal lumen and absorbed by duodenal mucosal cells [5]. Retinyl esters and provitamin A carotenoids are converted to retinol after uptake into the lumen (for retinyl esters) or absorption (for provitamin A carotenoids). Retinol is then oxidized to retinal and retinoic acid, the two main active vitamin A metabolites in the body [1]. Most of the body's vitamin A is stored in the liver in the form of retinyl esters [1].

Retinol and carotenoid levels are typically measured in plasma or serum because blood samples are easy to collect [1]. However, these levels are not always reliable indicators of vitamin A status because they do not decline until vitamin A levels in the liver and other storage sites are almost depleted and because acute and chronic infections can decrease serum and plasma retinol concentrations [1]. Most vitamin A is stored in the liver, so measuring vitamin A levels in the liver is the best way to assess vitamin A adequacy [1]. In clinical studies, specialized research laboratories can measure liver vitamin A reserves indirectly using isotope-dilution or dose-response methods, in which plasma levels of retinol, a tracer surrogate, or both are measured over several days after the administration of vitamin A [1].

In clinical practice, plasma retinol levels alone can be used to document significant deficiency. A serum or plasma retinol concentration of 20 mcg/dL (0.70 micromoles/L) or less frequently reflects moderate vitamin A deficiency, and a level of 10 mcg/dL (0.35 micromoles/L) or less is considered an indicator of severe vitamin A deficiency [1].

Recommended Intakes

Intake recommendations for vitamin A and other nutrients are provided in the Dietary Reference Intakes (DRIs) developed by the Food and Nutrition Board (FNB) at the National Academies of Sciences, Engineering, and Medicine [5]. DRI is the general term for a set of reference values used for planning and assessing nutrient intakes of healthy people. These values, which vary by age and sex, include the following:

- Recommended Dietary Allowance (RDA): Average daily level of intake sufficient to meet the nutrient requirements of nearly all (97%–98%) healthy individuals; often used to plan nutritionally adequate diets for individuals
- Adequate Intake (AI): Intake at this level is assumed to ensure nutritional adequacy; established when evidence is insufficient to develop an RDA
- Estimated Average Requirement (EAR): Average daily level of intake estimated to meet the requirements of 50% of healthy individuals; usually used to assess the nutrient intakes of groups of people and to plan nutritionally adequate diets for them; can also be used to assess the nutrient intakes of individuals
- Tolerable Upper Intake Level (UL): Maximum daily intake unlikely to cause adverse health effects

RDAs for vitamin A are given as retinol activity equivalents (RAE) to account for the different bioactivities of retinol and provitamin A carotenoids, all of which are converted by the body into retinol (see Table 1). One mcg RAE is equivalent to 1 mcg retinol, 2 mcg supplemental beta-carotene, 12 mcg dietary beta-carotene, or 24 mcg dietary alpha-carotene or beta-cryptoxanthin [5].

Table 1: Recommended Dietary Allowances (RDAs) for Vitamin A [5]

Age	Male	Female	Pregnancy	Lactation
Birth to 6 months*	400 mcg RAE	400 mcg RAE		
7–12 months*	500 mcg RAE	500 mcg RAE		
1–3 years	300 mcg RAE	300 mcg RAE		
4–8 years	400 mcg RAE	400 mcg RAE		
9–13 years	600 mcg RAE	600 mcg RAE		
14–18 years	900 mcg RAE	700 mcg RAE	750 mcg RAE	1,200 mcg RAE
19–50 years	900 mcg RAE	700 mcg RAE	770 mcg RAE	1,300 mcg RAE
51+ years	900 mcg RAE	700 mcg RAE		

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*AI, equivalent to the mean intake of vitamin A in healthy, breastfed infants.

The units of measurement for vitamin A are now mcg RAE, but International Units (IUs) were previously used [10]. To convert IU to mcg RAE, use the following [11–13]:

- 1 IU retinol = 0.3 mcg RAE
- 1 IU supplemental beta-carotene = 0.3 mcg RAE
- 1 IU dietary beta-carotene = 0.05 mcg RAE
- 1 IU dietary alpha-carotene or beta-cryptoxanthin = 0.025 mcg RAE

RAE can only be directly converted into IUs if the sources of vitamin A are known. For example, the RDA of 900 mcg RAE for adolescent and adult men is equivalent to 3,000 IU if the food or supplement source is preformed vitamin A (retinol) or if the supplement source is beta-carotene. This RDA is also equivalent to 18,000 IU beta-carotene from food or to 36,000 IU alpha-carotene or beta-cryptoxanthin from food. Therefore, a mixed diet containing 900 mcg RAE provides between 3,000 and 36,000 IU vitamin A, depending on the foods consumed.

Sources of Vitamin A

Food

Concentrations of preformed vitamin A are highest in liver, fish, eggs, and dairy products [1]. Most dietary provitamin A in the U.S. diet comes from leafy green vegetables, orange and yellow vegetables, tomato products, fruits, and some vegetable oils [1,5,14]. Vitamin A is routinely added to some foods, including milk and margarine [1,2]. Some ready-to-eat cereals are also fortified with vitamin A.

About 65% to 80% of vitamin A consumed in the United States and other high-income countries comes from preformed vitamin A, whereas provitamin A is the main form consumed in low-income countries, where diets include more plant-based foods [2,15]. Among U.S. children and adolescents, enriched and fortified foods account for 34%–40% of vitamin A intakes from food [16].

The body might absorb up to 75% to 100% of retinol and, in most cases, 10% to 30% of beta-carotene from foods [17,18]. Cooking and heat treatment can increase the bioavailability of beta-carotene from foods [19].

Table 2 lists a variety of foods and their vitamin A content per serving. The foods from animal sources in Table 2 contain primarily preformed vitamin A, the plant-

based foods have provitamin A, and the foods with a mixture of ingredients from animals and plants contain both preformed vitamin A and provitamin A.

Table 2: Vitamin A Content of Selected Foods [20]

Food	microgram (mcg) RAE per serving	Percent DV*
Beef liver, pan fried, 3 ounces	6,582	731
Sweet potato, baked in skin, 1 whole	1,403	156
Spinach, frozen, boiled, ½ cup	573	64
Pumpkin pie, commercially prepared, 1 piece	488	54
Carrots, raw, ½ cup	459	51
Herring, Atlantic, pickled, 3 ounces	219	24
Ice cream, French vanilla, soft serve, ⅔ cup	185	21
Milk, skim, with added vitamin A and vitamin D, 1 cup	149	17
Cantaloupe, raw, ½ cup	135	15
Cheese, ricotta, part skim, ½ cup	133	15
Peppers, sweet, red, raw, ½ cup	117	13
Mangos, raw, 1 whole	112	12
Breakfast cereals, fortified with 10% of the DV for vitamin A, 1 serving	90	10
Egg, hard boiled, 1 large	75	8
Black-eyed peas (cowpeas), boiled, 1 cup	66	7
Apricots, dried, sulfured, 5 apricots	63	7
Broccoli, boiled, ½ cup	60	7
Salmon, sockeye, cooked, 3 ounces	59	7
Tomato juice, canned, ¾ cup	42	5
Yogurt, plain, low fat, 1 cup	32	4
Tuna, light, canned in oil, drained, 3 ounces	20	2
Baked beans, canned, plain or vegetarian, 1 cup	13	1
Summer squash, all varieties, boiled, ½ cup	10	1
Chicken, breast meat and skin, roasted, ½ breast	5	1
Pistachio nuts, dry roasted, 1 ounce	4	0

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*DV = Daily Value. Food and Drug Administration (FDA) developed DVs to help consumers compare the nutrient contents of foods and dietary supplements within the context of a total diet. The DV for vitamin A is 900 mcg RAE for adults and children age 4 years and older [11], where 1 mcg RAE = 1 mcg retinol, 2 mcg beta-carotene from supplements, 12 mcg beta-carotene from foods, 24 mcg alpha-carotene, or 24 mcg beta-cryptoxanthin. FDA does not require food labels to list vitamin A content unless vitamin A has been added to the food. Foods providing 20% or more of the DV are considered to be high sources of a nutrient, but foods providing lower percentages of the DV also contribute to a healthful diet.

The U.S. Department of Agriculture's (USDA's) [FoodData Central](https://fdc.nal.usda.gov/) (<https://fdc.nal.usda.gov/>) lists the nutrient content of many foods and provides a comprehensive list of foods containing vitamin A arranged by [nutrient content](https://ods.od.nih.gov/pubs/usdandb/VitaminA-Content.pdf) (<https://ods.od.nih.gov/pubs/usdandb/VitaminA-Content.pdf>) and by [food name](https://ods.od.nih.gov/pubs/usdandb/VitaminA-Food.pdf) (<https://ods.od.nih.gov/pubs/usdandb/VitaminA-Food.pdf>).

Dietary supplements

Vitamin A is available in stand-alone supplements and most multivitamins, often in the form of retinyl acetate, retinyl palmitate, provitamin A beta-carotene, or a combination [1,21]. Amounts of vitamin A in supplements vary widely, but 3,000 mcg RAE (333% of the DV) is common [21]. Multivitamins commonly have somewhat lower amounts, often 750 to 1,050 mcg RAE (83% to 117% of the DV).

The absorption of preformed vitamin A esters from dietary supplements is 70%–90%, and that of beta-carotene ranges from 8.7% to 65% [19,22].

Vitamin A Intakes and Status

Average daily intakes of vitamin A from foods and beverages in the United States were 682 mcg RAE for men age 20 and older and 616 mcg RAE for women in 2017–2018, according to the National Health and Nutrition Examination Survey (NHANES) [23]. For children age 2–19, mean daily intakes of vitamin A from foods and beverages ranged from 497 to 680 mcg RAE. An analysis of biochemical data from 2003–2006 NHANES data indicates that less than 1% of the U.S. population

has a serum retinol level of less than 20 mcg/dL, which indicates that vitamin A deficiency is uncommon in the U.S. population [24].

Data from NHANES III, conducted from 1988 to 1994, showed that approximately 26% of the vitamin A in RAEs consumed by men and 34% of that consumed by women in the United States comes from provitamin A carotenoids [5]. The remainder comes from preformed vitamin A, mostly in the form of retinyl esters.

About 12% to 40% of the U.S. population, depending on age, uses supplements containing vitamin A [25]. Adults age 71 years or older and children younger than 9 are more likely than members of other age groups to take supplements containing vitamin A.

Vitamin A Deficiency

Frank vitamin A deficiency is rare in the United States. However, vitamin A deficiency is still common in many developing countries, often as a result of limited access to foods containing preformed vitamin A from animal-based food sources and to foods containing provitamin A carotenoids because of poverty or traditional diets [1,26]. A pooled analysis of population-based surveys from 138 low-income and middle-income countries found that 29% of children age 6 months to 5 years had vitamin A deficiency in 2013 [27]. Deficiency rates were highest in sub-Saharan Africa (48%) and South Asia (44%). In addition, approximately 10% to 20% of pregnant women in low-income countries have vitamin A deficiency [28].

Limited research suggests that vitamin A deficiency may also be influenced by genetic variability in conversion rates of beta-carotene to vitamin A. Certain polymorphisms in the *BCMO1* gene have been found to reduce the activity of the BCMO1 enzyme in humans [8,9], and a study in the Philippines among 693 children and adolescents found an inverse association between vitamin A status and the A379V TT variant in the *BCMO1* gene [7].

The most common clinical sign of vitamin A deficiency is xerophthalmia, which develops after plasma retinol has been low and the eye's vitamin A reserves have become depleted. The first sign is night blindness, or the inability to see in low light or darkness as a result of low rhodopsin levels in the retina [1,27,28].

Xerophthalmia also affects the cornea and can eventually lead to permanent blindness; vitamin A deficiency is one of the top causes of preventable blindness in children [28].

Chronic vitamin A deficiency has also been associated with abnormal lung development, respiratory diseases (such as pneumonia), and an increased risk of anemia and death [26,27,29].

Another effect of chronic vitamin A deficiency is increased severity and mortality risk of infections (particularly measles and infection-associated diarrhea) [26]. In 2013, 94,500 children in low-income and middle-income countries died of diarrhea and 11,200 died of measles as a result of vitamin A deficiency [27]. More than 95% of deaths attributable to vitamin A deficiency occurred in sub-Saharan Africa and Asia, where vitamin A deficiency was responsible for 2% of all deaths in children younger than 5 years [27].

Groups at Risk of Vitamin A Inadequacy

The following groups are among those most likely to have inadequate intakes of vitamin A.

Premature infants

Preterm infants have low liver stores of vitamin A at birth, and their plasma concentrations of retinol often remain low throughout the first year of life [30,31]. Preterm infants with vitamin A deficiency have a higher risk of eye and chronic lung diseases [32,33]. However, in high-income countries, clinical vitamin A deficiency is rare in infants and occurs only in those with malabsorption disorders [34].

Infants, children, and pregnant and lactating women in low-income and middle-income countries

Pregnant women need extra vitamin A for fetal growth and tissue maintenance and to support their own metabolism [35-37]. The breast milk of lactating women with adequate vitamin A intakes contains sufficient amounts of vitamin A to meet infants' needs for the first 6 months of life [38]. However, in people with vitamin A

deficiency, the vitamin A content of breast milk is not sufficient to maintain adequate vitamin A stores in infants who are exclusively breastfed [38].

About 190 million preschool-age children (one-third of all children in this age group), mostly in Africa and Southeast Asia, have vitamin A deficiency, according to the World Health Organization [27,39]. They have a higher risk of visual impairment and of illness and death from childhood infections, such as measles and infections that cause diarrheal diseases [1,39].

The World Health Organization estimates that 9.8 million pregnant women (15% of all pregnant women) around the world, mostly in low-income and middle-income countries, have xerophthalmia as a result of vitamin A deficiency [40].

People with cystic fibrosis

Up to 90% of people with cystic fibrosis have pancreatic insufficiency, which increases their risk of vitamin A deficiency due to difficulty absorbing fat [1,41]. Studies in Australia and the Netherlands indicate that 2% to 13% of children and adolescents with cystic fibrosis have vitamin A deficiency [42,43]. As a result, standard care for cystic fibrosis includes lifelong treatment with vitamin A (daily amounts of 750 mcg RAE to 3,000 mcg RAE, depending on age, are recommended in the United States and Australia), other fat-soluble vitamins, and pancreatic enzymes [41,43].

Individuals with gastrointestinal disorders

Approximately one-quarter of children with Crohn's disease and ulcerative colitis have vitamin A deficiency; adults with these disorders, especially those who have had the disorder for several years, also have a higher risk of vitamin A deficiency [44,45]. Although some evidence supports the use of vitamin A supplements in people with these disorders [46], other research has found that supplementation offers no benefit [47]. Some children and adults with newly diagnosed celiac disease also have vitamin A deficiency; a gluten-free diet can, but does not always, eliminate this deficiency [48-51].

Vitamin A and Health

This section focuses on three diseases and disorders in which vitamin A or carotenoids might play a role: cancer, age-related macular degeneration (AMD), and measles.

Cancer

Because of its role in regulating cell growth and differentiation, several studies have examined the association between vitamin A and various types of cancer. However, the relationship between serum vitamin A levels or vitamin A supplementation and cancer risk or cancer-related death is unclear. This fact sheet does not include studies of all-trans retinoic acid, a vitamin A metabolite that is used as a drug in high doses to treat a form of leukemia [52,53].

Several systematic reviews and meta-analyses of observational studies have shown that higher dietary intakes of retinol, carotenoids, fruits and vegetables, or a combination are associated with a lower risk of lung cancer [54], non-Hodgkin lymphoma [55], pancreatic cancer [56], oral cavity cancer [57], laryngeal cancer [57], esophageal cancer [58], ovarian cancer [59,60], glioma [61], and bladder cancer [62]. However, other observational studies have found no association between intakes of different forms of vitamin A and risk of liver cancer [63], non-Hodgkin lymphoma [64], colorectal cancer [65], prostate cancer [65], or all cancers [66].

Some clinical trial evidence suggests that supplemental vitamin A might reduce the risk of certain cancers but increase the risk of other forms of cancer, cardiovascular disease morbidity and mortality, and all-cause mortality. Examples are provided below.

The Carotene and Retinol Efficacy Trial (CARET) included 18,314 male and female current and former smokers (with at least a 20 pack-year history [equivalent to smoking 1 pack per day for 20 years or 2 packs per day for 10 years, for example] of cigarette smoking) as well as some men occupationally exposed to asbestos (who also have a higher risk of lung cancer), all age 45–74 years. The study randomized participants to take supplements containing 30 mg beta-carotene plus 25,000 IU (7,500 mcg RAE) retinyl palmitate or a placebo daily for about 6 years to evaluate the potential effects on lung cancer risk [67]. The trial was ended prematurely after a mean of 4 years, partly because the supplements were

unexpectedly found to have increased lung cancer risk by 28% and death from lung cancer by 46%; the supplements also increased the risk of all-cause mortality by 17%.

A subsequent study followed CARET participants for an additional 6 years after they stopped taking the study supplements [68]. During this time, the differences in lung cancer risk between the intervention and placebo groups were no longer statistically significant, with one exception: women in the intervention group had a 33% higher risk of lung cancer. In a separate analysis of CARET study data, men who took the two supplements had a 35% lower risk of nonaggressive prostate cancer during the 4-year active trial but not during the 6-year postintervention period. In contrast, men who took these two supplements in addition to another self-prescribed supplement (typically a multivitamin) had a 52% higher risk of aggressive prostate cancer during the active trial, but not during the postintervention period [69].

The Alpha-Tocopherol, Beta-Carotene (ATBC) Cancer Prevention Study also found that beta-carotene supplements increased the risk of lung cancer in smokers [70]. In this study, 29,133 male smokers age 50–69 years who smoked an average of 20.4 cigarettes a day for an average of 35.9 years took a supplement containing 50 mg/day alpha-tocopherol, 20 mg/day beta-carotene, both alpha-tocopherol and beta-carotene, or a placebo for 5–8 years. The beta-carotene supplements increased the risk of lung cancer by 18%, although they had little to no effect on the incidence of other cancers. The overall rate of death, primarily from lung cancer and ischemic heart disease, was 8% higher in participants who took beta-carotene. A subsequent study followed 25,563 of these participants for an additional 18 years [71]. During this period, participants were no longer taking the supplements, but most continued to smoke. Participants who had taken beta-carotene in the original trial did not have a higher risk of lung cancer, but they had a 20% higher risk of death due to prostate cancer.

The Age-Related Eye Disease Study 2 (AREDS2) was a 5-year randomized clinical trial with 4,203 participants age 50–85 years examining the effects on AMD of a dietary supplement containing several ingredients with or without beta-carotene (15 mg [7,500 mcg RAE]) [72]. No current smokers received the supplements containing beta-carotene. At the end of the trial, more lung cancers were

discovered in the beta-carotene group than in the no beta-carotene group (23 vs. 11 cases), and 31 of the 34 affected were former smokers. In a follow-up analysis of 3,882 of the participants 5 years after the end of AREDS2 (during which they took the AREDS2 formulation containing lutein and zeaxanthin instead of beta-carotene), the increased lung cancer risk persisted, with an 82% higher risk among participants who took the supplement containing beta-carotene during the 5-year AREDS2 trial [73].

Three other clinical trials have found no relationship between taking vitamin A or beta-carotene supplements and lung cancer incidence or mortality [74]. One trial randomized 22,071 male physicians age 40–84 years to take 50 mg beta carotene on alternate days or a placebo for 12 years [75]. Eleven percent of the physicians were current smokers, and 38% were former smokers at the start of the study. The results showed no differences between the groups in number of cases of lung cancer or any malignant neoplasms or number of deaths from cancer. Another trial randomized 7,627 women (mean age 60.4 years) to take 50 mg beta-carotene on alternate days, 600 IU vitamin E on alternate days, 500 mg vitamin C daily, or a placebo for a mean of 9.4 years [76]. Fifteen percent of the women were current smokers, and 41% were former smokers at the start of the study. None of the supplements had any significant effect on total cancer incidence or cancer mortality, including from lung cancer. A third trial included 29,584 healthy men and women age 40–69 years who were living in Linxian, China, where micronutrient deficiencies are common [77]. The study randomized participants to take either a placebo or one of four vitamin and mineral combinations (including one providing retinol and zinc and another providing beta carotene, vitamin E, and selenium) for 5.25 years. The investigators followed participants for an additional 10 years after they stopped taking the supplements. The nutrient doses in the supplements were equivalent to or twice as high as U.S. recommended intakes, but the study report did not provide the exact doses. During both the intervention and follow-up periods, lung cancer death rates did not differ among the five groups, even when the investigators further analyzed the results for differences by age, sex, and smoking status.

The CARET and ATBC study results suggest that large supplemental doses of beta-carotene with or without retinyl palmitate have detrimental effects in current or former smokers and workers exposed to asbestos. However, the other studies

described above that used similar vitamin A doses but had smaller proportions of current or former smokers do not raise this concern. Among nonsmokers, beta-carotene and vitamin A supplements do not appear to affect the risk of cancer.

Age-related macular degeneration

AMD is the leading cause of significant vision loss in older people [78]. AMD's etiology involves complex interactions among genetic susceptibility, environmental factors (including exposure to oxidative stress), and normal aging [78]. Because of the role of oxidative stress in AMD pathophysiology, supplements containing carotenoids with antioxidant functions, such as beta-carotene, lutein, and zeaxanthin, might be useful for preventing or treating this condition. Lutein and zeaxanthin (which are not precursors of vitamin A), in particular, accumulate in the retina, the tissue in the eye that is damaged by AMD.

The AREDS trial found that participants with a high risk of developing advanced AMD (i.e., those who had intermediate AMD or who had advanced AMD in one eye) had a 25% lower risk of developing advanced AMD after they took a daily supplement containing beta-carotene (15 mg [7,500 mcg RAE]), vitamin E (180 mg [400 IU] dl-alpha-tocopheryl acetate), vitamin C (500 mg), zinc (80 mg), and copper (2 mg) for 5 years than participants taking a placebo [79].

The follow-up AREDS2 study confirmed the value of this supplement in reducing the progression of AMD over a median follow-up period of 5 years [72]. However, this follow-up study showed that adding lutein (10 mg) and zeaxanthin (2 mg) or omega-3 fatty acids to the formulation produced no additional benefits. Importantly, the follow-up study also revealed that beta-carotene was not a required ingredient; the original AREDS formulation without beta-carotene provided the same protective effect against developing advanced AMD.

In a more detailed analysis, participants with the lowest dietary intakes of lutein and zeaxanthin had a 26% lower risk of advanced AMD when they took a supplement containing these two carotenoids than those who did not take a supplement with these carotenoids [72]. The risk of advanced AMD was also 18% lower in participants who took the modified AREDS supplement containing lutein

and zeaxanthin but not beta-carotene than in participants who took the formulation with beta-carotene but not lutein or zeaxanthin.

A subsequent study monitored dietary intakes of several nutrients in 4,504 AREDS participants and 3,738 AREDS2 participants (mean age 71 years) for a median of 10.2 years [80]. Participants in the two highest quintiles of intakes for vitamin A, beta-carotene, or lutein and zeaxanthin had a lower risk of progression to late AMD. For example, the risk of late AMD was 18% lower among those in the fifth quintile for vitamin A intake and 20% lower among those in the fourth quintile than among those in the first quintile.

At the end of the 5-year AREDS2 trial, participants were all offered the final AREDS2 formulation that included lutein and zeaxanthin in place of beta-carotene.

Researchers followed up with 3,882 of these participants for an additional 5 years [73]. After 10 years, participants who had taken the AREDS2 supplement with lutein and zeaxanthin had an additional 20% reduced risk of progression to late AMD compared with those who took the supplement containing beta-carotene. This finding confirmed the benefit of replacing beta-carotene with lutein and zeaxanthin.

Individuals who have or are developing AMD should talk to their health care provider about their vitamin A intakes and the supplement formulations used in the AREDS studies.

Measles

In 2023, measles was responsible for about 107,500 deaths worldwide, mostly in young children in low-income countries [81,82]. Vitamin A deficiency, which is rare in the United States, is a risk factor for severe measles [27,83]. A World Health Organization analysis of data from 83 countries showed that 11,200 child deaths from measles were associated with vitamin A deficiency in 2013, and more than 95% of these deaths occurred in sub-Saharan Africa and south Asia [27].

Research suggests that vitamin A supplementation reduces the risk of measles in children who are at high risk of vitamin A deficiency. However, vitamin A supplementation does not appear to reduce the risk of death from measles. A 2022 Cochrane Review included six clinical trials that examined the effect of vitamin A supplementation on the risk of measles in children [84]. These studies enrolled a

total of 19,566 children age 6 months to 5 years who lived in low- and middle-income countries. Vitamin A doses ranged from 15,000 mcg RAE (50,000 IU) to 60,000 mcg RAE (200,000 IU), depending on age. Supplements were administered as a single dose or every 4 to 6 months. Vitamin A supplementation reduced the risk of new cases of measles by 50%. However, the same Cochrane Review found that vitamin A supplementation did not affect the risk of death due to measles, based on the results from six clinical trials in a total of 1,088,261 children.

Health Risks from Excessive Vitamin A

Because vitamin A is fat soluble, the body stores excess amounts, primarily in the liver, and these levels can accumulate.

Acute vitamin A toxicity, also referred to as hypervitaminosis A, occurs within days to weeks after someone ingests one or a few very high doses (typically more than 100 times the RDA) [85]. Resulting signs and symptoms typically include severe headache, blurred vision, nausea, dizziness, aching muscles, and coordination problems. In severe cases, cerebral spinal fluid pressure can increase, leading to drowsiness and, eventually, coma and even death [85].

Chronic hypervitaminosis A (regular consumption of high doses) can cause dry skin, painful muscles and joints, fatigue, depression, and abnormal liver test results [85].

Total intakes of preformed vitamin A that exceed the UL as well as some retinoid medications used as topical therapies (such as isotretinoin, used to treat severe acne, and etretinate, a treatment for severe psoriasis) can cause congenital birth defects [1]. These birth defects can include malformations of the eye, skull, lungs, and heart [14]. Experts advise women who are or might be pregnant and those who are lactating not to take high doses (more than 3,000 mcg RAE [10,000 IU] daily) of vitamin A supplements [1].

Unlike preformed vitamin A, beta-carotene is not known to be teratogenic or lead to reproductive toxicity [1]. The most common effect of long-term, excess beta-carotene is carotenoderma, a harmless condition in which the skin becomes yellow-orange [3]. This condition can be reversed by discontinuing beta-carotene

ingestion. However, the ATBC trial found that supplementation with a large amount of beta-carotene (20 mg/day), with or without 50 mg/day vitamin E, for 5–8 years increased the risk of lung cancer and mortality (mainly from lung cancer and ischemic heart disease) in male smokers [70]. The CARET trial also showed that supplementation with a large amount of beta-carotene (30 mg/day) plus 7,500 mcg RAE (25,000 IU)/day retinyl palmitate for 4–8 years in current and former smokers as well as some men occupationally exposed to asbestos increased the risk of lung cancer and death from lung cancer [67].

The FNB has not established ULs for beta-carotene and other provitamin A carotenoids [3]. However, the FNB advises against the use of beta-carotene supplements for the general population, except as a provitamin A source to prevent vitamin A deficiency.

Tolerable upper intake levels for preformed vitamin A

The FNB has established ULs for preformed vitamin A that apply to both food and supplement intakes [5]. The FNB based these ULs on the amounts associated with an increased risk of liver abnormalities in men and women, teratogenic effects, and several toxic effects in infants and children.

Table 3: Tolerable Upper Intake Levels (ULs) for Preformed Vitamin A [5]*

Age	Male	Female	Pregnancy	Lactation
Birth to 12 months	600 mcg	600 mcg		
1–3 years	600 mcg	600 mcg		
4–8 years	900 mcg	900 mcg		
9–13 years	1,700 mcg	1,700 mcg		
14–18 years	2,800 mcg	2,800 mcg	2,800 mcg	2,800 mcg
19+ years	3,000 mcg	3,000 mcg	3,000 mcg	3,000 mcg

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*These ULs apply only to products from animal sources and supplements whose vitamin A comes entirely from retinol or its ester forms, such as retinyl palmitate. However, many dietary supplements (such as multivitamins) do not provide all of their vitamin A in retinol or its ester forms. For example, the vitamin A in some supplements consists partly or entirely of beta-carotene. In such cases, the

percentage of retinol or retinyl ester in the supplement should be used to determine whether an individual's vitamin A intake exceeds the UL. For example, a supplement whose label indicates that the product contains 3,000 mcg RAE vitamin A and that 60% of this vitamin A comes from beta-carotene (and therefore 40% comes from retinol or retinyl ester) provides 1,200 mcg RAE of preformed vitamin A. That amount is above the UL for children from birth to 8 years but below the UL for older children and adults.

Interactions with Medications

Vitamin A has the potential to interact with certain medications. In addition, several types of medications might adversely affect vitamin A levels. A few examples are provided below. Individuals taking these and other medications on a regular basis should discuss their vitamin A status with their health care providers.

Orlistat (Alli, Xenical), a weight-loss treatment, can decrease the absorption of vitamin A, other fat-soluble vitamins, and beta-carotene, resulting in low plasma levels in some patients [86,87]. The manufacturers of Alli and Xenical recommend that patients on orlistat take a multivitamin supplement containing vitamin A and beta-carotene as well as other fat-soluble vitamins [88,89].

Retinoids

Several synthetic retinoids derived from vitamin A are used orally as prescription medicines. Examples include the psoriasis treatment acitretin (Soriatane) and bexarotene (Targretin), used to treat the skin effects of T-cell lymphoma. Retinoids can increase the risk of hypervitaminosis A when taken in combination with vitamin A supplements [87].

Vitamin A and Healthful Diets

In general, a person's nutritional needs should be met primarily through the diet, including fortified foods. Dietary supplements may be useful in cases where it is not possible to meet the needs for specific nutrients through food alone, especially during certain life stages. The *Dietary Guidelines for Americans* offers a general description of healthy dietary patterns.

For more information about building a healthy dietary pattern, refer to the *Dietary Guidelines for Americans* (<https://www.dietaryguidelines.gov/>).

References

1. Blaner WS. Vitamin A and Provitamin A Carotenoids. In: Marriott BP, Birt DF, Stallings VA, Yates AA, eds. *Present Knowledge in Nutrition*. 11th ed. Cambridge, Massachusetts: Wiley-Blackwell; 2020:73-91.
2. Ross A. Vitamin A. In: Ross AC, Caballero B, Cousins RJ, Tucker KL, Ziegler TR, eds. *Modern Nutrition in Health and Disease*. 11th ed. Baltimore, MD: Lippincott Williams & Wilkins; 2014:260-77.
3. Institute of Medicine. Food and Nutrition Board. Dietary Reference Intakes for Vitamin C, Vitamin E, Selenium, and Carotenoids (<https://www.nap.edu/read/9810/chapter/1>). Washington, DC: National Academy Press; 2000.
4. Carazo A, Macakova K, Matousova K, Krcmova LK, Protti M, Mladenka P. Vitamin A update: Forms, sources, kinetics, detection, function, deficiency, therapeutic use and toxicity. *Nutrients* 2021;13:1703. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/34069881/>)]
5. Institute of Medicine. Food and Nutrition Board. Dietary Reference Intakes for Vitamin A, Vitamin K, Arsenic, Boron, Chromium, Copper, Iodine, Iron, Manganese, Molybdenum, Nickel, Silicon, Vanadium, and Zinc (<https://www.nap.edu/read/10026/chapter/1>). Washington, DC: National Academy Press; 2001.
6. Kedishvili NY. Enzymology of retinoic acid biosynthesis and degradation. *J Lipid Res*. 2013 Jul;54(7):1744-60. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/23630397/>)]
7. Zumaraga MPP, Arquiza JMRA, Concepcion MA, Perlas L, Alcudia-Catalma MN, Rodriguez M. Genotype Effects on β -Carotene Conversion to Vitamin A: Implications on Reducing Vitamin A Deficiency in the Philippines. *Food Nutr Bull*. 2022 Mar;43(1):25-34. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/34903070/>)]
8. Lietz G, Oxley A, Leung W, Hesketh J. Single nucleotide polymorphisms upstream from the β -carotene 15,15'-monooxygenase gene influence provitamin A conversion efficiency in female volunteers. *J Nutr*. 2012

Jan;142(1):161S-5S. [PubMed abstract
(<https://pubmed.ncbi.nlm.nih.gov/22113863/>)]

9. Leung WC, Hessel S, Méplan C, Flint J, Oberhauser V, Tourniaire F, Hesketh JE, von Lintig J, Lietz G. Two common single nucleotide polymorphisms in the gene encoding beta-carotene 15,15'-monooxygenase alter beta-carotene metabolism in female volunteers. FASEB J. 2009 Apr;23(4):1041-53. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/19103647/>)]
10. U.S. Food and Drug Administration. Food Labeling: Revision of the Nutrition and Supplement Facts Labels (<https://www.federalregister.gov/documents/2016/05/27/2016-11867/food-labeling-revision-of-the-nutrition-and-supplement-facts-labels>). 2016.
11. U.S. Food and Drug Administration. Food Labeling: Revision of the Nutrition and Supplement Facts Labels and Serving Sizes of Foods That Can Reasonably Be Consumed at One Eating Occasion; Dual-Column Labeling; Updating, Modifying, and Establishing Certain Reference Amounts Customarily Consumed; Serving Size for Breath Mints; and Technical Amendments; Proposed Extension of Compliance Dates (<https://www.federalregister.gov/documents/2017/10/02/2017-21019/food-labeling-revision-of-the-nutrition-and-supplement-facts-labels-and-serving-sizes-of-foods-that>). 2017.
12. U.S. Food and Drug Administration. Converting Units of Measure for Folate, Niacin, and Vitamins A, D, and E on the Nutrition and Supplement Facts Labels: Guidance for Industry (<https://www.fda.gov/media/129863/download>). 2019.
13. Dwyer J, Saldanha L, Haggans C, Potischman N, Gahche J, Thomas P, et al. Conversions of β -carotene as vitamin A in IU to vitamin A in RAE. J Nutr 2020;150:1337. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/32367133/>)]
14. Solomons NW. Vitamin A. In: Bowman B, Russell R, eds. Present Knowledge in Nutrition. 9th ed. Washington, DC: International Life Sciences Institute; 2006:157-83.
15. Van Loo-Bouwman CA, Naber TH, Schaafsma G. A review of vitamin A equivalency of beta-carotene in various food matrices for human consumption. Br J Nutr 2014;111:2153-66. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/24513222/>)]
16. Berner LA, Keast DR, Bailey RL, Dwyer JT. Fortified foods are major contributors to nutrient intakes in diets of US children and adolescents. J Acad

- Nutr Diet 2014;114:1009-22.e8. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/24462266/\)](#)]
17. Reboul E. Absorption of vitamin A and carotenoids by the enterocyte: focus on transport proteins. *Nutrients* 2013;5:3563-81. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/24036530/\)](#)]
 18. Reboul E. Mechanisms of carotenoid intestinal absorption: where do we stand? *Nutrients* 2019;11:838. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/31013870/\)](#)]
 19. Tanumihardjo SA, Russell RM, Stephensen CB, Gannon BM, Craft NE, Haskell MJ, et al. Biomarkers of Nutrition for Development (BOND)-Vitamin A review. *J Nutr* 2016;146:1816S-48S. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/27511929/\)](#)]
 20. U.S. Department of Agriculture. [FoodData Central \(https://fdc.nal.usda.gov/\)](#). 2021.
 21. Office of Dietary Supplements, National Institutes of Health. [Dietary Supplement Label Database \(https://dslid.od.nih.gov/dslid/index.jsp\)](#). 2021.
 22. Haskell MJ. The challenge to reach nutritional adequacy for vitamin A: beta-carotene bioavailability and conversion--evidence in humans. *Am J Clin Nutr* 2012;96:1193s-203s. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/23053560/\)](#)]
 23. U.S. Department of Agriculture, Agricultural Research Service. [What We Eat in America, 2017-2018 \(https://www.ars.usda.gov/northeast-area/beltsville-md-bhnrc/beltsville-human-nutrition-research-center/food-surveys-research-group/docs/wweia-data-tables/\)](#). 2021.
 24. Bird JK, Murphy RA, Ciappio ED, McBurney MI. Risk of deficiency in multiple concurrent micronutrients in children and adults in the United States. *Nutrients* 2017;9: 655. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/28672791/\)](#)]
 25. Bailey RL, Gahche JJ, Lentino CV, Dwyer JT, Engel JS, Thomas PR, et al. Dietary supplement use in the United States, 2003-2006. *J Nutr* 2011;141:261-6. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/21178089/\)](#)]
 26. Wiseman EM, Bar-El Dadon S, Reifen R. The vicious cycle of vitamin a deficiency: A review. *Crit Rev Food Sci Nutr* 2017;57:3703-14. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/2712815/\)](#)]
 27. Stevens GA, Bennett JE, Hennocq Q, Lu Y, De-Regil LM, Rogers L, et al. Trends and mortality effects of vitamin A deficiency in children in 138 low-income and

middle-income countries between 1991 and 2013: a pooled analysis of population-based surveys. *Lancet Glob Health* 2015;3:e528-36. [[PubMed abstract](https://pubmed.ncbi.nlm.nih.gov/26275329/) (https://pubmed.ncbi.nlm.nih.gov/26275329/)]

28. Bailey RL, West KP, Jr., Black RE. The epidemiology of global micronutrient deficiencies. *Ann Nutr Metab* 2015;66 Suppl 2:22-33. [[PubMed abstract](https://pubmed.ncbi.nlm.nih.gov/26045325/) (https://pubmed.ncbi.nlm.nih.gov/26045325/)]
29. Timoneda J, Rodriguez-Fernandez L, Zaragoza R, Marin MP, Cabezuelo MT, Torres L, et al. Vitamin A deficiency and the lung. *Nutrients* 2018;10:1132. [[PubMed abstract](https://pubmed.ncbi.nlm.nih.gov/30134568/) (https://pubmed.ncbi.nlm.nih.gov/30134568/)]
30. Schwartz E, Zelig R, Parker A, Johnson S. Vitamin A supplementation for the prevention of bronchopulmonary dysplasia in preterm infants: an update. *Nutr Clin Pract* 2017;32:346-53. [[PubMed abstract](https://pubmed.ncbi.nlm.nih.gov/28537511/) (https://pubmed.ncbi.nlm.nih.gov/28537511/)]
31. Darlow BA, Graham PJ. Vitamin A supplementation to prevent mortality and short and long-term morbidity in very low birthweight infants. *Cochrane Database Syst Rev* 2007:CD000501. [[PubMed abstract](https://pubmed.ncbi.nlm.nih.gov/21975731/) (https://pubmed.ncbi.nlm.nih.gov/21975731/)]
32. Rakshasbhuvankar A, Patole S, Simmer K, Pillow JJ. Enteral vitamin A for reducing severity of bronchopulmonary dysplasia in extremely preterm infants: a randomised controlled trial. *BMC Pediatr* 2017;17:204. [[PubMed abstract](https://pubmed.ncbi.nlm.nih.gov/29246130/) (https://pubmed.ncbi.nlm.nih.gov/29246130/)]
33. Sun H, Cheng R, Wang Z. Early vitamin A supplementation improves the outcome of retinopathy of prematurity in extremely premature infants. *Retina* 2020;40:1176-84. [[PubMed abstract](https://pubmed.ncbi.nlm.nih.gov/30964778/) (https://pubmed.ncbi.nlm.nih.gov/30964778/)]
34. Mactier H, Weaver LT. Vitamin A and preterm infants: what we know, what we don't know, and what we need to know. *Arch Dis Child Fetal Neonatal Ed* 2005;90:F103-8. [[PubMed abstract](https://pubmed.ncbi.nlm.nih.gov/15724031/) (https://pubmed.ncbi.nlm.nih.gov/15724031/)]
35. McCauley ME, van den Broek N, Dou L, Othman M. Vitamin A supplementation during pregnancy for maternal and newborn outcomes. *Cochrane Database Syst Rev* 2015:Cd008666. [[PubMed abstract](https://pubmed.ncbi.nlm.nih.gov/26503498/) (https://pubmed.ncbi.nlm.nih.gov/26503498/)]
36. Haider BA, Sharma R, Bhutta ZA. Neonatal vitamin A supplementation for the prevention of mortality and morbidity in term neonates in low and middle income countries. *Cochrane Database Syst Rev* 2017;2:Cd006980. [[PubMed abstract](https://pubmed.ncbi.nlm.nih.gov/28234402/) (https://pubmed.ncbi.nlm.nih.gov/28234402/)]

37. Ota E, da Silva Lopes K, Middleton P, Flenady V, Wariki WM, Rahman MO, et al. Antenatal interventions for preventing stillbirth, fetal loss and perinatal death: an overview of Cochrane systematic reviews. *Cochrane Database Syst Rev* 2020;12:Cd009599. [[PubMed abstract](#) (<https://pubmed.ncbi.nlm.nih.gov/33336827/>)]
38. Oliveira JM, Allert R, East CE. Vitamin A supplementation for postpartum women. *Cochrane Database Syst Rev* 2016;3:Cd005944. [[PubMed abstract](#) (<https://pubmed.ncbi.nlm.nih.gov/27012320/>)]
39. World Health Organization. Guideline: Vitamin A Supplementation in Infants and Children 6-59 Months of Age (<https://www.who.int/publications/i/item/9789241501767>). 2011.
40. World Health Organization. Global Prevalence of Vitamin A Deficiency in Populations at Risk 1995–2005: WHO Global Database on Vitamin A Deficiency. Geneva: World Health Organization; 2009.
41. de Vries JJ, Chang AB, Bonifant CM, Shevill E, Marchant JM. Vitamin A and beta (beta)-carotene supplementation for cystic fibrosis. *Cochrane Database Syst Rev* 2018;8:Cd006751. [[PubMed abstract](#) (<https://pubmed.ncbi.nlm.nih.gov/30091146/>)]
42. Rana M, Wong-See D, Katz T, Gaskin K, Whitehead B, Jaffe A, et al. Fat-soluble vitamin deficiency in children and adolescents with cystic fibrosis. *J Clin Pathol* 2014;67:605-8. [[PubMed abstract](#) (<https://pubmed.ncbi.nlm.nih.gov/24711511/>)]
43. Woestenenk JW, Broos N, Stellato RK, Arets HG, van der Ent CK, Houwen RH. Vitamin A intake and serum retinol levels in children and adolescents with cystic fibrosis. *Clin Nutr* 2016;35:654-9. [[PubMed abstract](#) (<https://pubmed.ncbi.nlm.nih.gov/25975494/>)]
44. Fabisiak N, Fabisiak A, Watala C, Fichna J. Fat-soluble vitamin deficiencies and inflammatory bowel disease: systematic review and meta-analysis. *J Clin Gastroenterol* 2017;51:878-89. [[PubMed abstract](#) (<https://pubmed.ncbi.nlm.nih.gov/28858940/>)]
45. Rempel J, Grover K, El-Matary W. Micronutrient deficiencies and anemia in children with inflammatory bowel disease. *Nutrients* 2021;13:236. [[PubMed abstract](#) (<https://pubmed.ncbi.nlm.nih.gov/33467587/>)]
46. Santucci NR, Alkhouri RH, Baker RD, Baker SS. Vitamin and zinc status pretreatment and posttreatment in patients with inflammatory bowel disease.

J Pediatr Gastroenterol Nutr 2014;59:455-7. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/25000354/\)](#)]

47. Wright JP, Mee AS, Parfitt A, Marks IN, Burns DG, Sherman M, et al. Vitamin A therapy in patients with Crohn's disease. Gastroenterology 1985;88:512-4. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/3965341/\)](#)]
48. Tokgöz Y, Terlemez S, Karul A. Fat soluble vitamin levels in children with newly diagnosed celiac disease, a case control study. BMC Pediatr 2018;18:130. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/29631542/\)](#)]
49. Wierdsma NJ, van Bokhorst-de van der Schueren MAE, Berkenpas M, Mulder CJJ, van Bodegraven AA. Vitamin and mineral deficiencies are highly prevalent in newly diagnosed celiac disease patients. Nutrients 2013;5:3975-92. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/24084055/\)](#)]
50. McGrogan L, Mackinder M, Stefanowicz F, Aroutiounova M, Catchpole A, Wadsworth J, et al. Micronutrient deficiencies in children with coeliac disease; a double-edged sword of both untreated disease and treatment with gluten-free diet. Clin Nutr 2021;40:2784-90. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/33933744/\)](#)]
51. Shepherd SJ, Gibson PR. Nutritional inadequacies of the gluten-free diet in both recently-diagnosed and long-term patients with coeliac disease. J Hum Nutr Diet 2013;26:349-58. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/23198728/\)](#)]
52. Wang ZY, Chen Z. Acute promyelocytic leukemia: from highly fatal to highly curable. Blood 2008;111:2505-15. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/18299451/\)](#)]
53. Lo-Coco F, Avvisati G, Vignetti M, Thiede C, Orlando SM, Iacobelli S, et al. Retinoic acid and arsenic trioxide for acute promyelocytic leukemia. N Engl J Med 2013;369:111-21. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/23841729/\)](#)]
54. Rowles JL, 3rd, Ranard KM, Smith JW, An R, Erdman JW, Jr. Increased dietary and circulating lycopene are associated with reduced prostate cancer risk: a systematic review and meta-analysis. Prostate Cancer Prostatic Dis 2017;20:361-77. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/28440323/\)](#)]
55. Chen F, Hu J, Liu P, Li J, Wei Z, Liu P. Carotenoid intake and risk of non-Hodgkin lymphoma: a systematic review and dose-response meta-analysis of

observational studies. *Ann Hematol* 2017;96:957-65. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/28011986/\)](https://pubmed.ncbi.nlm.nih.gov/28011986/)]

56. Chen J, Jiang W, Shao L, Zhong D, Wu Y, Cai J. Association between intake of antioxidants and pancreatic cancer risk: a meta-analysis. *International Journal of Food Sciences and Nutrition* 2016;67:744-53. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/27356952/\)](https://pubmed.ncbi.nlm.nih.gov/27356952/)]
57. Leoncini E, Nedovic D, Panic N, Pastorino R, Edefonti V, Boccia S. Carotenoid intake from natural sources and head and neck cancer: a systematic review and meta-analysis of epidemiological studies. *Cancer Epidemiol Biomarkers Prev* 2015;24:1003-11. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/25873578/\)](https://pubmed.ncbi.nlm.nih.gov/25873578/)]
58. Li H, He P, Lin T, Guo H, Li Y, Song Y, et al. Association between plasma retinol levels and the risk of all-cause mortality in general hypertensive patients: A nested case-control study. *J Clin Hypertens (Greenwich)* 2020;22:906-13. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/32352642/\)](https://pubmed.ncbi.nlm.nih.gov/32352642/)]
59. Li X, Xu J. Meta-analysis of the association between dietary lycopene intake and ovarian cancer risk in postmenopausal women. *Scientific reports* 2014;4:4885. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/24810584/\)](https://pubmed.ncbi.nlm.nih.gov/24810584/)]
60. Wang Q, He C. Dietary vitamin A intake and the risk of ovarian cancer: a meta-analysis. *Biosci Rep* 2020;40. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/32149329/\)](https://pubmed.ncbi.nlm.nih.gov/32149329/)]
61. Lv W, Zhong X, Xu L, Han W. Association between dietary vitamin A Intake and the risk of glioma: evidence from a meta-analysis. *Nutrients* 2015;7:8897-904. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/26516909/\)](https://pubmed.ncbi.nlm.nih.gov/26516909/)]
62. Tang JE, Wang RJ, Zhong H, Yu B, Chen Y. Vitamin A and risk of bladder cancer: a meta-analysis of epidemiological studies. *World J Surg Oncol* 2014;12:130. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/24773914/\)](https://pubmed.ncbi.nlm.nih.gov/24773914/)]
63. Leelakanok N, D'Cunha RR, Sutamtewagul G, Schweizer ML. A systematic review and meta-analysis of the association between vitamin A intake, serum vitamin A, and risk of liver cancer. *Nutr Health* 2018;24:121-31. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/29792083/\)](https://pubmed.ncbi.nlm.nih.gov/29792083/)]
64. Psaltopoulou T, Ntanasis-Stathopoulos I, Tsilimigras DI, Tzanninis IG, Gavriatopoulou M, Sergentanis TN. Micronutrient intake and risk of hematological malignancies in adults: a systematic review and meta-analysis

of cohort studies. Nutr Cancer 2018;70:821-39. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/30288994/\)](https://pubmed.ncbi.nlm.nih.gov/30288994/)]

65. Wang X, Yang H-H, Liu Y, Zhou Q, Chen Z-H. Lycopene consumption and risk of colorectal cancer: a meta-analysis of observational studies. Nutrition and Cancer 2016;68:1083-96. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/27472298/\)](https://pubmed.ncbi.nlm.nih.gov/27472298/)]
66. Aune D, Keum N, Giovannucci E, Fadnes LT, Boffetta P, Greenwood DC, et al. Dietary intake and blood concentrations of antioxidants and the risk of cardiovascular disease, total cancer, and all-cause mortality: a systematic review and dose-response meta-analysis of prospective studies. Am J Clin Nutr 2018;108:1069-91. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/30475962/\)](https://pubmed.ncbi.nlm.nih.gov/30475962/)]
67. Omenn GS, Goodman GE, Thornquist MD, Balmes J, Cullen MR, Glass A, et al. Effects of a combination of beta carotene and vitamin A on lung cancer and cardiovascular disease. N Engl J Med 1996;334:1150-5. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/8602180/\)](https://pubmed.ncbi.nlm.nih.gov/8602180/)]
68. Goodman GE, Thornquist MD, Balmes J, Cullen MR, Meyskens FL, Omenn GS, et al. The beta-carotene and retinol efficacy trial: incidence of lung cancer and cardiovascular disease mortality during 6-year follow-up after stopping β -carotene and retinol supplements. J Natl Cancer Inst 2004;96:1743-50. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/15572756/\)](https://pubmed.ncbi.nlm.nih.gov/15572756/)]
69. Neuhouwer ML, Barnett MJ, Kristal AR, Ambrosone CB, King IB, Thornquist M, et al. Dietary supplement use and prostate cancer risk in the Carotene and Retinol Efficacy Trial. Cancer Epidemiol Biomarkers Prev 2009;18:2202-6. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/19661078/\)](https://pubmed.ncbi.nlm.nih.gov/19661078/)]
70. The Alpha-Tocopherol, Beta-Carotene Cancer Prevention Study Group. The effect of vitamin E and beta carotene on the incidence of lung cancer and other cancers in male smokers. N Engl J Med 1994;330:1029-35. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/8127329/\)](https://pubmed.ncbi.nlm.nih.gov/8127329/)]
71. Virtamo J, Taylor PR, Kontto J, Männistö S, Utriainen M, Weinstein SJ, et al. Effects of α -tocopherol and β -carotene supplementation on cancer incidence and mortality: 18-year postintervention follow-up of the Alpha-tocopherol, Beta-carotene Cancer Prevention Study. International journal of cancer 2014;135:178-85. [[PubMed abstract \(https://pubmed.ncbi.nlm.nih.gov/24338499/\)](https://pubmed.ncbi.nlm.nih.gov/24338499/)]

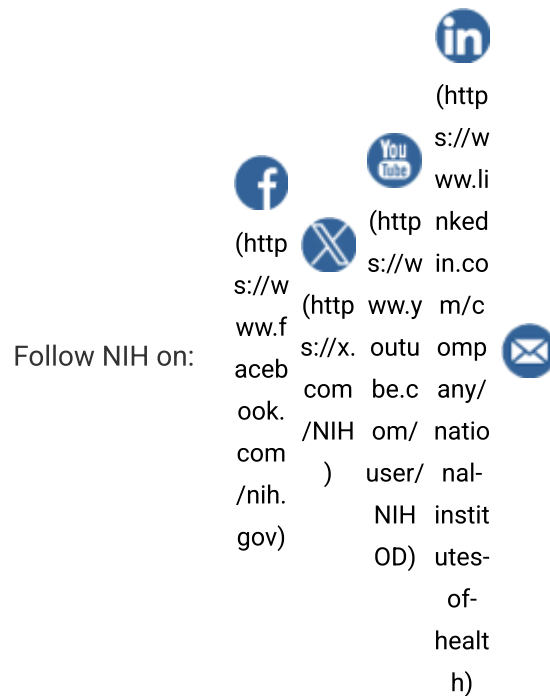
72. Age-Related Eye Disease Study 2 Research Group. Lutein + zeaxanthin and omega-3 fatty acids for age-related macular degeneration: the Age-Related Eye Disease Study 2 (AREDS2) randomized clinical trial. JAMA 2013;309:2005-15. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/23644932/>)]
73. Chew EY, Clemons TE, Agron E, Domalpally A, Keenan TDL, Vitale S, Weber C, Smith DC, Christen W, for the AREDS2 Research group. Long-term outcomes of adding Lutein/Zeaxanthin and Omega-3 Fatty Acids to the AREDS Supplements on Age-Related Macular Degeneration Progression: AREDS2 Report #28. JAMA Ophthalmology. June 2, 2022. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/35653117/>)]
74. Cortes-Jofre M, Rueda JR, Asenjo-Lobos C, Madrid E, Bonfill Cosp X. Drugs for preventing lung cancer in healthy people. Cochrane Database Syst Rev 2020;3:Cd002141. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/32130738/>)]
75. Hennekens CH, Buring JE, Manson JE, Stampfer M, Rosner B, Cook NR, et al. Lack of effect of long-term supplementation with beta carotene on the incidence of malignant neoplasms and cardiovascular disease. The New England journal of medicine 1996;334:1145-9. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/8602179/>)]
76. Lin J, Cook NR, Albert C, Zaharris E, Gaziano JM, Van Denburgh M, et al. Vitamins C and E and beta carotene supplementation and cancer risk: a randomized controlled trial. J Natl Cancer Inst 2009;101:14-23. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/19116389/>)]
77. Kamangar F, Qiao YL, Yu B, Sun XD, Abnet CC, Fan JH, et al. Lung cancer chemoprevention: a randomized, double-blind trial in Linxian, China. Cancer Epidemiol Biomarkers Prev 2006;15:1562-4. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/16896051/>)]
78. Fleckenstein M, Keenan TDL, Guymer RH, Chakravarthy U, Schmitz-Valckenberg S, Klaver CC, et al. Age-related macular degeneration. Nat Rev Dis Primers 2021;7:31. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/33958600/>)]
79. Age-Related Eye Disease Study Research Group. A randomized, placebo-controlled, clinical trial of high-dose supplementation with vitamins C and E, beta carotene, and zinc for age-related macular degeneration and vision loss:

- AREDS report no. 8. Arch Ophthalmol 2001;119:1417-36. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/11594942/>)]
80. Agron E, Mares J, Clemons TE, Swaroop A, Chew EY, Keenan TDL. Dietary nutrient intake and progression to late age-related macular degeneration in the Age-Related Eye Disease Studies 1 and 2. Ophthalmology 2021;128:425-42. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/32858063/>)]
81. Minta AA, Ferrari M, Antoni S, Lambert B, Sayi TS, et al. Progress Toward Measles Elimination - Worldwide, 2000-2023. MMWR Morb Mortal Wkly Rep 2024;73:1036-42. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/39541251/>)]
82. World Health Organization. Measles (<https://www.who.int/news-room/fact-sheets/detail/measles>). 2024.
83. Haggerty PA. Medical Nutrition Therapy for Infectious Diseases. In: Raymond JL, Morrow K, eds. Krause and Mahan's Food and the Nutrition Care Process, 16th Edition: Elsevier; 2022.
84. Imdad A, Mayo-Wilson E, Haykal MR, Regan A, Sidhu J, et al. Vitamin A supplementation for preventing morbidity and mortality in children from six months to five years of age. Cochrane Database Syst Rev 2022;3:Cd008524. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/35294044/>)]
85. National Institute of Diabetes and Digestive and Kidney Diseases. LiverTox: Clinical and research information on drug-induced liver injury (<https://www.ncbi.nlm.nih.gov/books/NBK548165/>). 2020.
86. Zhi J, Melia AT, Koss-Twardy SG, Arora S, Patel IH. The effect of orlistat, an inhibitor of dietary fat absorption, on the pharmacokinetics of beta-carotene in healthy volunteers. J Clin Pharmacol 1996;36:152-9. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/8852391/>)]
87. Ulbricht C, Basch E, Chao W, Conquer J, Costa D, Culwell S, et al. An evidence-based systematic review of vitamin A by the natural standard research collaboration. J Diet Suppl 2012;9:299-416. [PubMed abstract (<https://pubmed.ncbi.nlm.nih.gov/23157584/>)]
88. Genentech USA, Inc. Xenical Package Insert (https://xenical.com/pdf/PI_Xenical-brand_FINAL.PDF). 2017.
89. GlaxoSmithKline. Alli Package Insert (<https://www.myalli.com/content/dam/cf-consumer-healthcare/myalli/products/alli%20120%20ct%20product%20label%20drug%20facts.pdf>). 2017.

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